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First American Corporation (Pvt.) Ltd.

Pakistan ACCORD
for Health and Safety in the Garment and Textile Industry

19 Km Main Daska Road, Ghuinke
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Latitude: [32.42408], longitude: [74.46865]
Date of inspection: 09-Mar-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 09-03-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 09-Mar-2026, Sarah Aiman and Azka Riaz on behalf of the Pakistan Accord carried out an initial electrical inspection for First American Corporation (Pvt.) Ltd., a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with Imran Malik (Director), Khalid Malik (Director Operations), Hamid Mustafa (Civil Engineer), Muhammad Awais (Surveyor) and other staff members of First American Corporation Pvt Ltd. The assessment was conducted as part of an overall Structural, Fire, and Electrical initial inspection.

The factory operates in single shift. Approximately 1400 workers operate in the factory with 20 workers being females.

The factory consists of 16 structures owned by First American Corporation Pvt Ltd. During the inspection, the areas highlighted in blue in Figure 1.1 were surveyed, with the site boundary indicated in red dashed line.

- **Admin Block** is a two storey (Ground + 1) building. The building is used for offices (Ground Floor), offices, sleeping areas (First Floor) and solar panels (Roof).
- **Garment Building** is a four storey (Ground + 3) building. The building is used for storage (Ground Floor), stitching hall (First and Second Floor), screen printing area, storage and solar panels (Third Floor).
- **Belt Stitching and Packing Department Building** is a three storey (Ground + 2) building. The building is used for packing area (Ground Floor), garments packing (First Floor), belt stitching (Second Floor).
- **Hummel Building** is a three story (Ground + 2) building. The building is used for cutting, packing, office (Ground Floor), stitching (First Floor), finishing and packing (Second Floor).
- **Heat Transfer / Bag Production Building** is a two storey (Ground + 1) building. The building is used for heat transfer department (Ground Floor), bag stitching (First Floor) and solar panels (Roof).
- **Pad Store / Bag Packing / Embroidery Building** is a two storey (Ground + 1) building. The building is used for pad store (Ground Floor), screen printing (First Floor) and solar panels (Roof).
- **Canteen / Sublimation Building** is a two storey (Ground + 1) building. The building is used for labour canteen (Ground Floor) and sublimation (First Floor).
- **Gloves Store and Guest Room Building** is a two storey (Ground + 1) building. The building is used for storage (Ground Floor), carton storage and guest room (First Floor).
- **Pad Production Hall** is a single storey (Ground only) building. The building is used for pad production (Ground Floor).
- **Belt / Accessories Production Building** is a two storey (Ground + 1) building. The building is used for belt production, gloves cutting (Ground Floor) and gloves stitching (First Floor).
- **HR Office** is a two storey (Ground + 1) building. The building is used for offices (Both Floors).
- **Mosque** is a two storey (Ground + 1) building. The building is used for prayer area (Both Floors).
- **Security Office** is a multiple single storey (Ground only) structures The structures are used for offices (Ground Floor).

• **Water Filtration Plant** is a single storey (Ground only) structure. The structure is used for water filtration (Ground Floor).
• **Pad Store Shed** is a single storey (Ground only) structure. The structure is used for storage (Ground Floor).
• **Generator Shed** is a single story (Ground only) structure. The structure is used for generator operations (Ground Floor).

A summary of the main concerns in relation to life safety systems identified during the inspection (non-exhaustive) that should be actioned is as follows:

• **Abnormal Temperature**: Within electrical panels, abnormal temperatures were recorded at various locations (refer to section 7.2 for further details). Hotspots within panels can cause short circuits and fire due to insulation damage and overheating of cables or electrical components.
• **Fire Alarm and Detection System**: Two conventional fire alarm control panels were installed in the facility. One panel was inoperational while other was under installation. This creates a major fire safety risk because the alarm system will not operate in case of fire, leading to delayed fire detection, delayed occupant warning, and delayed evacuation.
• **Emergency lighting System**: The emergency lighting system had inadequate coverage along evacuation routes, with non-functional fixtures, and non-illuminated directional signage in various areas of the facility, hindering safe evacuation during power outages and increasing the risk of accidents.
• **Earthing system**: The earthing system was found to be missing throughout the facility. No earth distribution was available in the electrical panels and machines. The lack of a proper earthing system poses significant safety risks, such as electric shocks, fire hazards, and potential equipment damage.

We see no reason to suspend operations in the facility due to these concerns, subject to the actions identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents

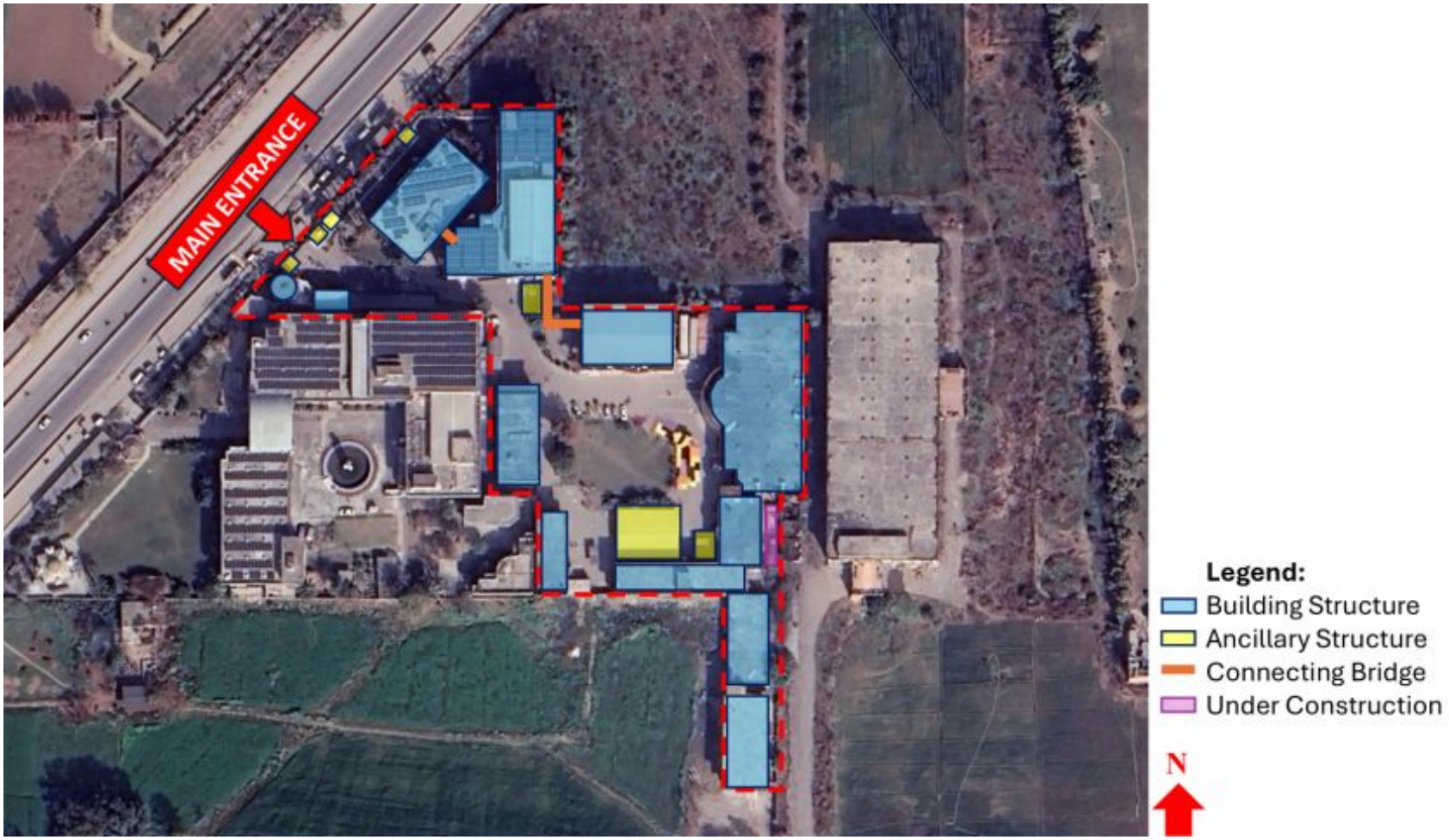


Figure 1.1. Site plan (Satellite view) – Red dashed line indicates site boundary, highlighted area (blue) indicates buildings inspected

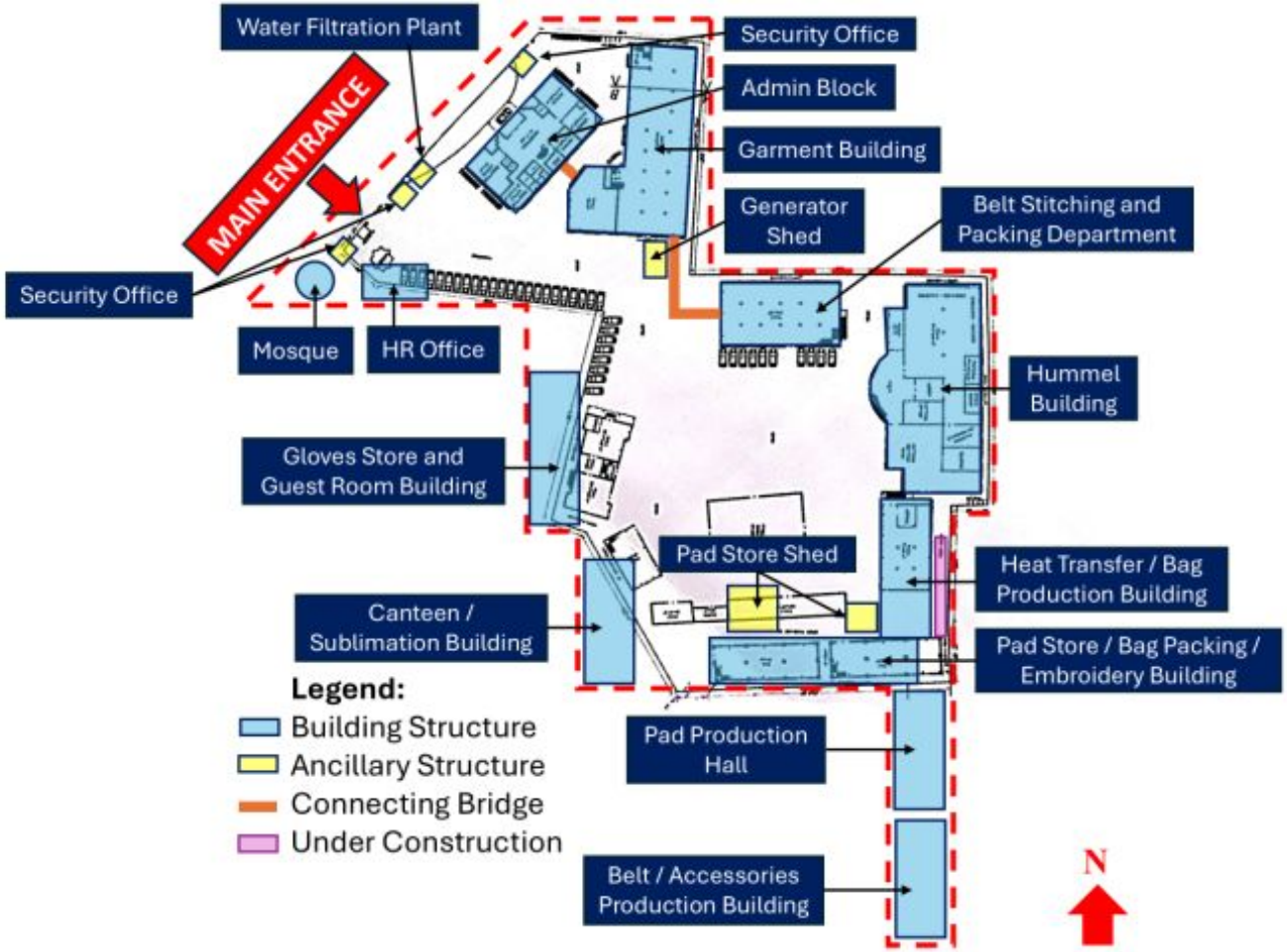


Figure 1.2. Site layout plan - Red dashed line indicates site boundary, highlighted area (blue) indicates buildings inspected

2.1 Building Use

The factory comprises twelve main buildings and four ancillary structures. The floor-wise use of buildings is as follows:

1) Admin Block
L00 – Ground Floor: Offices
L01 – First Floor: Offices and sleeping areas
L99 – Roof : Solar panels

2) Garments Building
L00 – Ground Floor: Storage
L01 – First Floor: Stitching Hall
L02 – Second Floor: Stitching Hall
L03 – Third Floor: Screen printing area, storage and solar panels

3) Belt Stitching and Packing Department Building
L00 – Ground Floor: Packing area
L01 – First Floor: Garments packing
L02 – Second Floor: Belt stitching

4) Hummel Building
L00 – Ground Floor: Cutting, packing, office
L01 – First Floor: Stitching
L02 – Second Floor: Finishing and packing

5) Heat Transfer / Bag Production Building
L00 – Ground Floor: Heat transfer department
L01 – First Floor: Bag stitching
L99 – Roof : Solar panels

6) Pad Store / Bag Packing / Embroidery Building
L00 – Ground Floor: Pad store
L01 – First Floor: Screen printing
L99 – Roof : Solar panels

7) Canteen / Sublimation Building
L00 – Ground Floor: Labour canteen
L01 – First Floor: Sublimation Hall

8) Gloves Store and Guest Room Building
L00 – Ground Floor: Storage
L01 – First Floor: Carton storage and guest room

9) Pad Production Hall

L00 – Ground Floor: Pad production

10) Belt / Accessories Production Building

L00 – Ground Floor: Belt production, gloves cutting

L01 – First Floor: Gloves stitching

11) HR Office

L00 – Ground Floor: Offices

L01 – First Floor: Offices

12) Mosque

L00 – Ground Floor: Prayer area

L01 – First Floor: Prayer area

Ancillary Structures

1) Security Office

L00 – Ground Floor: Offices

2) Water Filtration Plant

L00 – Ground Floor: Water filtration plant

3) Pad Store Shed

L00 – Ground Floor: Storage

4) Generator Shed

L00 – Ground Floor: Generator Area

2.2 Utility Source

The facility receives two independent 400 V utility feeders from Gujranwala Electric Power Company (GEPCO) via two GEPCO-owned step-down transformers (200 kVA, 200 kVA) located outside the main gate. The incoming power is routed to the changeover panels in the generator area. The electrical supply is then distributed throughout the facility from these changeover panels via cables. Additionally, a 330 kW solar system has been installed on-site and is synchronized with the facility's primary supply.

2.3 Backup source

The facility has two low-voltage diesel generators—one rated at 315 kVA, one at 250 kVA, located in Generator Areas, serving as backup power sources. The fire alarm and detection system, along with the emergency lighting system, are equipped with built-in battery backup supplies.

3. Documentation

This table shows the details of the documentation provided by the factory.

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	No		
2	Single Line Diagrams	No		
3	Fire Alarm Drawing	No		
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	N/A		

4. Electrical Safety Measures

During our electrical site inspection of First American Corporation (Pvt.) Ltd., a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within First American Corporation (Pvt.) Ltd.. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a licensed professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202603-063691

Timeline	Remediation Plan
Medium Term CAP-0074410	Factories should begin remediation of CAP items – such as cleaning, cable organization, and basic corrections – using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no electrical room layout drawings available for the factory.
There are no visible indications of electrical equipment locations or pathways for maintenance and emergency access.
The absence of electrical room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies.

This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202603-063692

Timeline	Remediation Plan
Medium Term CAP-0074411	Develop detailed electrical room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0074412	Regularly update and review the electrical room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.
The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel.Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202603-063694

Timeline	Remediation Plan
Medium Term CAP-0074415	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0074416	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202603-063693

Timeline	Remediation Plan
Medium Term CAP-0074413	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0074414	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey, the circuit breakers installed throughout the facility, inside the electrical panels and distribution boards, were inspected.

i. It was observed that the circuit breaker switch was broken in storage area distribution board in Garments Building. Broken circuit breaker switch makes emergency isolation difficult, posing fire, shock, and arc flash risks.

ii. It was observed at some locations that circuit breakers were damaged and burned out. Damaged and burned-out circuit breakers pose a serious fire hazard and risk of electrical shock due to failure to interrupt fault currents and accidental touch. This issue was specifically observed in the cutting distribution board on the ground floor of the Hummel Building.

iii. It was found that the circuit breaker was installed in outdoor areas without weather-protective enclosures, posing a risk of short circuits or electrical incidents during adverse weather conditions. This issue was specifically observed outside Generator Area, outside the Garments Building, and outside the Belt Stitching Building.

iv. It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes.
a. In the stitching hall of Garments Building, a 150 A circuit breaker was connected to 6 mm^2 PVC cables.
b. In the packing area of Belt stitching & Packing Building, a 100 A circuit breaker was connected to 4 mm^2 PVC cables and 6 mm^2 PVC cables.
c. In the garments packing area of Belt stitching & Packing Building, a 100 A circuit breaker was connected to 10 mm^2 PVC cables.
d. In the cutting & packing area, a distribution board on the ground floor of Hummel Building, a 200 A circuit breaker was connected to 25 mm^2 PVC cables, 225 A circuit breaker was connected to 16 mm^2 PVC cables, 100 A circuit breaker was connected to 16 mm^2 PVC cables and 150 A circuit breaker was connected to 35 mm^2 PVC cables.
e. In the lighting distribution board on the ground floor of Hummel Building, a 100 A circuit breaker was connected to 6 mm^2 PVC cables.
f. In the belt production area distribution board on the ground floor of Accessories Production Building, a 150 A circuit breaker was connected to 10 mm^2 PVC cables.
g. In the sublimation hall distribution board on the first floor of Sublimation Building, a 100 A circuit breaker was connected to 6 mm^2 PVC cables.
h. In the distribution board of Admin Block Building, a 100 A circuit breaker was connected to 6 mm^2 PVC cables.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202603-064025

Timeline	Remediation Plan
Immediate CAP-0076037	i. Replace broken circuit breaker immediately and test all distribution board breakers for proper operation.
Medium Term CAP-0076034	ii. Replace all damaged and burned-out circuit breakers with properly rated units and ensure proper installation and regular inspection to maintain safe operation.
Medium Term CAP-0076035	iii. Relocate circuit breaker boxes to indoor or sheltered areas. If relocation is not feasible, install weatherproof enclosures with a suitable IP rating for outdoor conditions, as per BS 7671 Clause 512.2.
Medium Term CAP-0076036	iv. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - 225 A and 100 A circuit breakers connected to 16 mm² PVC cables in cutting distribution board of Hummel Building



Figure 2.3 - Circuit breaker switch is broken in storage area distribution board of Garments Building



Figure 2.2 - Circuit breakers found damaged and burned out in the cutting distribution board of Hummel Building



Figure 2.4 - Circuit breaker installed outside Generator area without weather protected enclosure

7.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical distribution systems can be caused by several factors, including:

> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

It was also found that the thermographic report was not available at the site. Without thermography testing, faults such as overheating components or loose connections may go unnoticed, which increases the risk of fire. Early detection through thermography can prevent fire incidents.

The categories of identified hot spots have been evaluated as per the ANSI NETA MTS 2019 standard, Table 100.18. Hot spots were observed, particularly at the following locations:
i. 93.7°C at the circuit breaker on the first floor of Belt Stitching and Packing Building
ii. 79.4°C on cable connection in solar distribution board in Garments Building
iii. 70.4°C at the circuit breaker and 60°C on the blue cable connection on first floor of Hummel Building
iv. 62.2°C on the red cable connection on second floor of Belt Stitching and Packing Building

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC-insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature

event that can cause severe injuries or fatalities to personnel working near the affected equipment.F-202603-064035

Timeline	Remediation Plan
Immediate CAP-0075987	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term CAP-0075988	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0075989	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating.



Figure 2.5 - Hot spot found on the circuit breaker in Belt Stitching and Packing Building



Figure 2.6 - Hot spot found on the circuit breaker in Belt Stitching and Packing Building



Figure 2.7 - Hot spot found on the cable solar inverters distribution board in Garments Building



Figure 2.8 - Hot spot found on the cable solar inverters distribution board in Garments Building



Figure 2.9 - Hot spot found on the circuit breaker in distribution board of Hummel Building



Figure 2.10 - Hot spot found on the circuit breaker in distribution board of Hummel Building

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. No earthing system design documentation was provided by the factory, indicating a lack of verification of the earthing system’s compliance and ongoing performance in accordance with relevant standards.

ii. The earth pits were available in the facility, but they were inaccessible.

iii. The earth pit testing and maintenance records were not provided.

iv. The earthing system was almost absent in the overall facility. All distribution boards lacked earthing cables, and electrical machines did not have proper earth connections.

v. Equipotential bonding for the cable trays and metallic panel covers was missing. This issue was observed in all areas of the facility.

The lack of a proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202603-064027

Timeline	Remediation Plan
Medium Term CAP-0076045	i. Design and evaluate the facility's earthing system, and develop layout drawings in accordance with BS 7671 18th Edition + A2 (2022). Submit the earthing layout drawings to the Pakistan Accord for review.
Medium Term CAP-0076046	ii. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.
Medium Term CAP-0076047	iii. Implement periodic earth testing per BS 7671 Regulation 643.7.1 to verify grounding effectiveness and submit testing record to Pakistan Accord for review.

Medium Term CAP-0076048	iv. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Medium Term CAP-0076049	v. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Longer Term CAP-0076050	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.11 - Inaccessible earth pit near Generator Area



Figure 2.12 - Earth connection missing for the stitching machines



Figure 2.13 - Equipotential bonding and earth connection missing in hydraulic jack distribution board



Figure 2.14 - Earth connection missing for manual cutter machines in cutting area of Hummel Building



Figure 2.15 - Earth connection missing in the stitching distribution board in Hummel Building



Figure 2.16 - Earth connection missing in the distribution board of HR Office Building



Figure 2.17 - Earth connection missing in the lighting distribution board on ground floor of Hummel building

7.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.

i. Combustible materials were found inside and around electrical panels and distribution boards, including circuit breakers mounted on wooden supports within the enclosures. Such materials pose a serious fire risk in the event of electrical faults. This issue was noted in the generator area, Garments Building, Belt Stitching Building, and Gloves Store Building.

ii. Some electrical panels exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements. This issue was observed in the distribution boards located in the Generator area.

iii. It was observed on the first floor of Belt Stitching Building that electrical components were not suitable for the environmental conditions in which they were installed. Unsuitable electrical components exposed to harsh environmental conditions (dust, moisture, heat, vibration) can cause insulation breakdown, short circuits, overheating, corrosion, and electrical fires.

iv. Visible damage and burn marks were observed on several panels, along with deteriorated cable insulation. This indicates possible overheating, arcing, or prior fault conditions, which can compromise the distribution system's integrity and increase the risk of short circuits, fire, and electric shock. This issue was observed in distribution board of Hummel Building, Belt Stitching Building and Garments Building.

v. Phase separators were missing in the generator area's panel and distribution board on ground floor of Hummel Building, increasing the risk of accidental contact between phases.

vi. Inadequate cable dressing was observed in most of the distribution boards, with conductors not properly arranged, secured, or terminated neatly. This resulted in untidy wiring, poor segregation between circuits, and unnecessary strain on terminations, which can lead to loose connections, overheating, and difficulty during inspection or fault-finding. This issue was specifically observed in the Generator Area, distribution boards of Garments Building and the Hummel Building.

vii. Protective circuit breakers were not provided between the changeover panel and the transformer. Incoming and outgoing cables were connected directly to the busbars without protective devices, creating a significant risk of fire, fault escalation, and equipment damage.

The utilization of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202603-

Timeline	Remediation Plan
Medium Term CAP-0076038	i. Remove all combustible materials stored inside or in close proximity to electrical panels and equipment, and maintain clearances to prevent potential fire hazards. Also, replace combustible non-compliant enclosures with noncombustible or fire-rated manufacturer approved enclosures.
Medium Term CAP-0076039	ii. Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Medium Term CAP-0076040	iii. Identify and replace electrical panels, cables, and busbars that are not suitable for the environmental conditions in which they are installed in accordance with Section 512.2 of BS 7671 18th Edition + A2 (2022).
Medium Term CAP-0076041	iv. Inspect and repair or replace all damaged panels, burnt circuit breakers, damaged cable insulation, and faulty instruments to ensure the integrity of the electrical system, in accordance with Section 522 of BS 7671 18th Edition + A2 (2022).
Medium Term CAP-0076042	v. Phase separators should be installed between adjacent live conductors of different phases within the distribution boards to ensure safe segregation and protection against direct contact, in accordance with BS 7671 Regulation 416.2.
Medium Term CAP-0076043	vi. Ensure proper cable management by organizing and securing cables within the panels using cable ties, clips, or ducting. Verify that all components and connections are securely mounted, neatly arranged, and labelled to improve safety, ease of maintenance, and compliance with electrical standards such as BS 7671.
Medium Term CAP-0076843	vii. Install appropriate circuit breakers to ensure protection for all incoming and outgoing feeders, reducing the risk of fire, equipment failure, and electrical hazards.
Longer Term CAP-0076044	Establish a schedule for regular maintenance and inspection of the panel, cabling, and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.18 - Non IP rated panel at first floor of Belt Stitching and Packing Building



Figure 2.19 - Circuit breaker installed on combustible material in distribution board of garments packing



Figure 2.20 - Circuit breakers missing in the changover panel in Generator Area



Figure 2.21 - Corrosion observed in distribution board located in Generator area



Figure 2.22 - Inadequate cable dressing and phase separators were missing near main breaker in Generator Area



Figure 2.23 - Combustible material found near main breaker installed in Generator Area



Figure 2.24 - Sign of burning observed at sample stitching distribution board in Garments Building

7.5 Loop Connection

This relates to the entire facility.

During the inspection, loop connections were identified within the electrical panels. This issue was specifically observed in the distribution boards located at the following locations:
i. Ground floor and first floor of the Garments Building
ii. Ground floor and first floor of the Belt Stitching Building
iii. Ground floor of Hummel Building
iv. First floor of Pad Store Building
v. Pad Production Hall
vi. Ground floor of Belt Production Building
vii. Outside Generator Area

In these loop connections, one circuit breaker is energized by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202603-064030

Timeline	Remediation Plan
Medium Term CAP-0076063	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilized to provide each circuit with the required live, neutral, and earth connections.



Figure 2.25 - Loop connections found in the main distribution board outside Generator Area



Figure 2.26 - Loop connections found in the distribution board of belt production area



Figure 2.27 - Loop connections found in the pad production hall



Figure 2.28 - Loop connections found in the distribution board of printing area in Pad Store Building



Figure 2.29 - Loop connections found in the distribution board of belt stitching area

7.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility. This issue was specifically observed in the distribution boards located at
i) Ground floor and first floor of the Garments Building
ii) First floor and second floor of Hummel Building
iii) Ground floor of Pad Store and Embroidery Building
iv) First floor of Sublimation Building
v) Ground floor of the Gloves Store Building
vi) Ground floor of Accessories Production Building

In these cases, multiple circuits were being protected by a single circuit breaker.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202603-064023

Timeline	Remediation Plan
Medium Term CAP-0076030	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022).



Figure 2.30 - Multiple connections found in the stitching distribution board of Hummel Building



Figure 2.31 - Multiple connections found in the ground floor of Pad Store Building



Figure 2.32 - Multiple connections found in the distribution board of stitching hall, Garments Building



Figure 2.33 - Multiple connections found in the distribution board of first floor, Garments Building

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

Unsealed openings were observed in multiple distribution boards, including the hydraulic jack, lift panel, water filtration plant, ground floor of the Belt Stitching Building, Gate Office Building, and Generator Area. The inadequate sealing allowed lizard ingress in some panels, indicating poor enclosure integrity and creating a risk of unplanned shutdowns, electrical faults, fire hazards, equipment damage, and unsafe conditions for maintenance personnel during operation or fault conditions.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202603-064028

Timeline	Remediation Plan
Medium Term CAP-0076052	Seal all openings in the distribution boards and replace any missing covers or panels to restore enclosure integrity and prevent the entry of reptiles, rodents, and foreign objects. These corrective actions shall ensure the safety and integrity of the electrical system in compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0076053	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.34 - Lizard found in hydraulic jack distribution board



Figure 2.35 - Unsealed openings observed in garments packing distribution board



Figure 2.36 - Unsealed openings observed in change over panel in Generator Area



Figure 2.37 - Unsealed openings observed at distribution board of Gate Office Building



Figure 2.38 - Unsealed openings observed in generator panel

7.8 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At numerous locations, it was found that the electrical panels do not possess the required space for inspection or easy access to the panels and their components.

In several instances, the placement of electrical panels adjacent to machines that continuously reject heat has caused the surrounding temperature to exceed ambient levels by several degrees. This temperature rise further increases the temperature of the panels and boards nearby. This includes the distribution boards located at the following locations.

i. Stitching hall and sampling hall on the first floor of the Garments Building
ii. Garments office on the first floor of Garments Building
iii. Packing area on the ground floor of Belt Stitching Building
iv. Cutting & Packing on ground floor of Hummel Building
v. Water Filtration Plant
vi. Ground floor of HR Office Building
vii. Pad Store Shed

Limited accessibility to distribution panels and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202603-064029

Timeline	Remediation Plan
Medium Term CAP-0076059	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.39 - Inadequate accessibility to panels in Ped Store Shed



Figure 2.41 - Inadequate accessibility to distribution board at HR Office



Figure 2.40 - Inadequate accessibility to Water Filtration Plant panel



Figure 2.42 - Inadequate accessibility to distribution board in packing area of Belt Stitching Building

7.9 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards, and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels, and trunking systems. This includes the following locations
i. Screen printing area distribution board of the Garments Building
ii. Lift Panel
iii. Bag stitching hall distribution hall of the Heat Transfer Building
iv. Gate office distribution board

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202603-064024

Timeline	Remediation Plan
Medium Term CAP-0076032	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0076033	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.43 - Lint and dust found in the lift panel



Figure 2.44 - Lint and dust found in distribution board of screen printing area



Figure 2.45 - Lint and dust found in distribution board of gate office

7.10 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

It was observed that a lightning protection system was provided; however, there was no study available to confirm its adequacy, and no risk assessment was available.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202603-064032

Timeline	Remediation Plan
Medium Term CAP-0076064	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment according to BS EN 62305 (Protection against Lightning) standard. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0076065	If the risk assessment concludes that an LPS is necessary, then hire a qualified engineer to perform an adequacy study of the existing LPS. This study should evaluate whether the current system meets the necessary design and performance standards according to BS EN 62305. If the study finds that the current LPS is inadequate, make the necessary upgrades to bring the system up to standard.



Figure 2.46 - Lightning arrestor installed on the roof of Garments Building

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

The facility has two conventional fire alarm control panels: One was completely non-functional, and the other was not commissioned and under installation on the day of inspection. The installed Fire Alarm and Detection System was non-listed, non-functional, and exhibited several deficiencies, including inadequate placement of smoke detectors, inappropriate detection and manual call points (MCP) coverage, and non-fire-rated cables. System testing was not possible as both fire alarm panels were non-operational.

Non-listed panels may not provide reliable detection and alarm functions, which can compromise timely evacuation and firefighting efforts. A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm. A delayed response during a fire event can significantly increase the impact on personnel, goods, and property.F-202603-064033

Timeline	Remediation Plan
Immediate CAP-0076078	Ensure the functionality and performance of the existing alarm and detection system within the building.
Medium Term CAP-0076079	i. Submit design documentation for the new replacement FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0076080	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.

Longer Term CAP-0076081	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.
Longer Term CAP-0076082	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.
Longer Term CAP-0076083	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0076084	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.47 - Smoke detectors missing at first floor of Belt Stitching Building



Figure 2.48 - Manual call point blocked in Pad Production Hall Building



Figure 2.49 - Non operational fire alarm control panel observed at security office



Figure 2.50 - Non fire rated cables used for fire alarm system

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection, it was observed that emergency lighting and illuminated exit signs were provided but not functional in a number of locations, particularly in the
i) Ground floor, first floor, and rooftop of Garments Building
ii) Ground floor and second floor of Hummmel Building
iii) First floor of Pad Store Building
iv) Ground floor of Pad Production Building
v) First floor of the Gloves Store Building
vi) Admin Block

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202603-064037

Timeline	Remediation Plan
Medium Term CAP-0075917	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.

Longer Term CAP-0075918	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.51 - Non functional emergency light in Admin Block



Figure 2.52 - Non functional emergency light on first floor of Garments Building



Figure 2.53 - Non-functional emergency light and exit light found on first floor of Pad Store Building



Figure 2.54 - Non-functional emergency light and exit light found on ground floor in Hummel Building

9.2 Provided but insufficient

The section refers to the entire facility.

During the inspection the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. The layouts for emergency lighting were missing. The absence of emergency lighting layouts throughout the facility raises concerns regarding the adequacy of illumination provided during power failures or emergencies.

ii. Several areas lack sufficient emergency lighting and exit directional signs coverage. This issue was specifically observed at
a) stairs from first floor to second floor of the Belt Stitching and Packing Building
b) stairs of second floor of Hummel Building
c) ground floor of Heat Transfer Building
d) first floor stairs of Belt Production Building
e) Water Filtration Plant

iii. The installed emergency lights were too dim to provide adequate illumination. This issue was observed with the emergency light installed on the ground floor of the Garments Building.

iv. The backup power supply to the emergency luminaires was insufficient. The emergency light battery backup test failed in the Generator Area and on the first floor of the Garments Building.

v. External emergency lighting was not provided, particularly in the assembly area and external emergency stairs, preventing people from reaching a place of safety outside the facility during nighttime evacuations.

vi. The facility did not maintain inspection, testing, and maintenance records for emergency lighting and exit sign systems, which is a noncompliance with NFPA 101 requirements.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202603-064034

Timeline	Remediation Plan
Medium Term CAP-0075899	i. Emergency lighting layouts must be developed in accordance with the Building Code of Pakistan Fire Safety Provisions 2016 and submitted to Pakistan Accord for review.
Medium Term CAP-0075900	ii. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Install additional emergency luminaires in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation.
Medium Term CAP-0075901	iii. Replace dim emergency lights with higher intensity luminaires that meet the minimum illumination levels specified in the Building Code of Pakistan Fire Safety Provisions 2016. Ensure that all emergency lights provide adequate brightness to facilitate safe evacuation during emergencies.
Medium Term CAP-0075902	iv. Upgrade the backup power supply to ensure it can adequately support all emergency luminaires during a power failure. This includes installing additional battery packs or generators as necessary to comply with the Building Code of Pakistan Fire Safety Provisions 2016.
Medium Term CAP-0075903	v. Install external emergency luminaires to enable safe evacuation during hours of darkness, all the way to a safe place outside the buildings.
Medium Term CAP-0075904	vi. Maintain proper monthly and annual emergency lighting test records as required by NFPA 101, ensuring all inspections, functional tests, and corrective actions are documented and kept on-site.
Longer Term CAP-0075905	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.55 - Inadequate battery backup of emergency light at generator area



Figure 2.56 - External emergency luminaire not provided at assembly area



Figure 2.57 - Emergency lighting and exit directional signs not installed at Water Filtration Plant



Figure 2.58 - Emergency lighting and exit directional signs not installed on ground floor in Heat Transfer Building

9.3 Independent power source

During the inspection, it was observed that the emergency lighting within the facility was not connected to an independent dedicated circuit.

The circuitry of emergency lights and exit signs be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202603-064036

Timeline	Remediation Plan
Longer Term CAP-0075907	Reconfigure emergency lights for independent circuit. Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

9.4 General Lighting

The section refers to the entire facility.

During the inspection the general lighting throughout the facility was assessed.

The following observations were noted:

i. General lighting from the second floor to the rooftop of the Garment's Building was found to be inadequate, resulting in poor visibility for routine operations and safe access.

ii. General lighting was controlled directly from the breakers rather than through standard light switches in the Garments Building

Insufficient general lighting in several locations can increase the risk of accidents and impede safe evacuation during emergencies, while controlling general lighting via breakers instead of light switches poses operational inefficiencies and safety hazards. This setup can delay emergency responses and heighten the risk of electrical faults or fires due to frequent breaker use. Addressing these issues is crucial for enhancing everyday safety and ensuring effective emergency preparedness in the facility.F-202603-064031

Timeline	Remediation Plan
Medium Term CAP-0075923	i. Conduct a thorough assessment of lighting throughout the facility to find areas with insufficient illumination. Install additional lighting fixtures in these spots to ensure adequate brightness. Ensure the new lighting design meets safety and efficiency guidelines to provide clear visibility for all tasks and emergency situations.
Medium Term CAP-0075924	ii. Reconfigure the electrical control system to ensure that general lighting is operated via standard wall-mounted light switches instead of circuit breakers. This involves installing appropriate switches and rewiring the control circuits as per the National Electrical Code (NEC) and other relevant electrical safety standards. Ensure that all switches are easily accessible and clearly labelled to enhance operational efficiency and safety.



Figure 2.59 - General lighting missing on stairs of second floor



Figure 2.60 - Lights controlled by circuit breaker in Garments Building

10. Generator, Transformer & Sub-Station Room

10.1 Working clearance

This section refers to the electrical switchrooms, generator room(s).

During our inspection, the working clearances surrounding key electrical equipment were assessed.

During the inspection, it was observed that the working clearance was inadequate in multiple areas. Equipment was placed too close to walls or other machines, making it difficult for personnel to access for routine maintenance or emergency interventions. Additionally, some pathways were obstructed, restricting movement and potentially delaying response times during an emergency. This issue was specifically observed with the panels installed in the inverter room of the solar and electrical panels installed on ground floor of the Garments Storage Building.

Proper working clearance is crucial for ensuring life and fire safety. Adequate clearance allows personnel to safely access and maintain equipment, thereby reducing the risk of accidents or injuries caused by cramped conditions. Insufficient working clearance increases the risk of equipment failure due to limited maintenance access, which can lead to fires or other hazardous situations F-202603-064039

Timeline	Remediation Plan
Medium Term CAP-0075985	Reorganize the equipment layout to ensure adequate clearance, remove obstructions in pathways, and implement clear floor markings and signage to designate working areas and pathways.
Longer Term CAP-0075986	Conduct regular workspace audits to maintain clearances, update maintenance protocols to include clearance checks, provide personnel training on maintaining clearances, and consider installing adjustable equipment to accommodate flexible positioning.



Figure 2.61 - Inadequate working clearance for Electrical Panel installed on ground floor of Garments Building



Figure 2.62 - Inadequate working clearance for solar's inverter on ground floor of Garments Building

10.2 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

Battery terminals were exposed and not properly insulated, increasing the risk of electrical shock or short circuits. This issue was observed with the generator batteries placed in the generator area.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202603-064038

Timeline	Remediation Plan
Medium Term CAP-0075983	Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term CAP-0075984	Establish a routine inspection and maintenance protocol to regularly check and maintain proper insulation of all battery terminals.



Figure 2.63 - Exposed terminals observed on the generator battery

11. Cabling & Wiring

11.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

It was observed that proper supports were not provided for cables; they were laid directly on the ground and hung along the walls. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure. This issue was particularly observed in the Generator Area and in the cutting and stitching areas on the ground floor and first floor of the Hummel Building.

Unsupported cables laid on the floor or loosely hung on walls are vulnerable to mechanical damage (crushing, abrasion, impact) and moisture ingress, which can lead to insulation failure, short-circuiting, and electrical fire.F-202603-063698

Timeline	Remediation Plan
Medium Term CAP-0076100	Secure cables with appropriate supports and insulate any exposed connections to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations)
Longer Term CAP-0076101	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.



Figure 2.64 - Inadequate cable supports observed in Generator Area



Figure 2.65 - Cables for stitching machines laying on ground in Hummel Building



Figure 2.66 - Cables for manual cutters laying on ground in Hummel Building

11.2 Cable trench and channel cleanliness

This related to the whole facility.

During the inspection of the facility, it was observed that the cleanliness of the cable trunking system had not been maintained. Numerous trunking systems were found to be filled with a significant amount of fluff and dust, which poses a serious fire risk.

This issue was observed outside the Generator Area, bag stitching hall on first floor of Heat Transfer Building, ground and first floor of Garments Building.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimise fire risks associated with accumulated debris.F-202603-063699

Timeline	Remediation Plan
Medium Term CAP-0076129	Clean all cable trunking systems to remove accumulated dust and fluff, ensuring that any potential fire hazards are mitigated.
Longer Term CAP-0076130	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of dust and fluff, thereby maintaining a safe environment and complying with safety standards.



Figure 2.67 - Cable channel filled with lint and dust at storage area of Garments Building



Figure 2.68 - Cable trench filled with dust and debris outside the Generator Area

11.3 Splicing of cables

This relates to the whole facility.

During the inspection, the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in broken strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. This issue was observed in the distribution boards on ground and first floor of Hummel building, Admin Block, first floor of Garments Building, Sublimation Hall and the ground floor of Belt Stitching Building.

ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. This issue was specifically observed in the distribution boards located on first floor of Garments Building, first floor of Belt Stitching Building, Generator Area and Gate Office Building.

iii. Non-compliant PVC connectors used in distribution boards, which could pose a hazard of a potential short circuit due to their poor insulation. This issue was specifically observed in the distribution boards and electrical enclosures of the Water Filtration Plant, first floor and roof of Garments Building and on ground floor of Hummel Building.

iv. Open, exposed, and idle cable connections were found inside distribution boards, posing a risk of electric shock, short circuit, and fire. This issue was specifically observed in distribution boards located at Gate Office, ground floor of Belt Stitching Building and on ground floor of hummel Building.

For any future electrical work, the termination of stranded cables must be done with cable lugs, and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202603-063700

Timeline	Remediation Plan

Medium Term CAP-0076131	i. Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).
Medium Term CAP-0076154	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.
Medium Term CAP-0076155	iii. Replace the PVC connector with a standard quality terminal block in accordance with BS7671 18th Edition + A2 (2022).
Medium Term CAP-0076159	iv. Remove unused cables or properly terminate and insulate open and exposed connections to eliminate the risk of sparks and potential fires. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 416.1 for basic insulation of live parts).
Longer Term CAP-0076156	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.
Longer Term CAP-0076157	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Longer Term CAP-0076158	iii. Establish a plan to replace all low-quality PVC connectors with compliant electrical terminal blocks and integrate this into regular maintenance to ensure ongoing safety and compliance.
Longer Term CAP-0076160	iv. Establish routine inspections to ensure all electrical terminations remain properly insulated and protected from exposure, especially in high-risk operational areas.



Figure 2.69 - PVC connectors observed in distribution board of HR Office



Figure 2.70 - Cable termination without lugs in main distribution board outside Generator Area



Figure 2.71 - Tape joints and cables without lugs were observed in distribution board of Water Filtration Plant



Figure 2.72 - Idle cables and PVC connectors found in water pump distribution board in Generator Area

11.4 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can lead to incorrect connections, potentially creating unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires. These observations were made at various locations throughout the site, including but not limited to the following.

i. Ground floor, first floor and roof of the Garments Building
ii. First floor of Belt Stitching Building
iii. Hydraulic jack distribution board
iv. Water Filtration Plant distribution board
v. Gate Office distribution board
vi. First floor of Sublimation Building
vii. Lift Panel

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202603-063697

Timeline	Remediation Plan
Medium Term CAP-0076094	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.

Longer Term CAP-0076095	Envelop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them up to date on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.73 - Incorrect color coding observed in the screen printing distribution board in Garments Building



Figure 2.74 - Incorrect color coding observed in the distribution board of Water Filtration Plant



Figure 2.75 - Incorrect color coding observed in the distribution board pf garments packing in Belt Stitching Building



Figure 2.76 - Incorrect color coding observed in the stitching distribution board in Hummel Building

11.5 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

The following observations were noted:

i. Power receptacles installed in an inappropriate location observed in the steamer area outside Garments Building and Pad Production Building: Receptacles placed in unsuitable locations, such as non-IP-rated sockets installed outdoors or exposed to the elements, are at higher risk of moisture ingress and corrosion. This can lead to electrical short circuits or fires, posing a severe safety hazard.

ii. Power receptacles and bulbs mounted on combustible wooden ply in the Generator Area and Pad Production Building. Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety.

iii. Broken and exposed power receptacles were observed inGarments Building and Belt Stitching and Packing Building. Damaged receptacles with exposed parts can lead to loose connections and increased electrical resistance. This condition can cause overheating, potentially igniting surrounding materials and increasing the risk of electrical fires, posing a serious life safety threat.

iv. A power receptacle was found to be improperly installed on the UPS in the ground floor of Pad Production Building, posing a risk of electric shock, overheating, short circuiting, and potential fire due to restricted ventilation and possible mechanical damage to the cable.

v. A lighting bulb holder has been found outside screen printing area in Garments Building, inappropriately connected and improperly mounted on the wall without any secure fixture or back box enclosure. The associated wiring cables are unmounted, loosely hanging exposed from the wall cavity without any conduit, trunking, or cable management in place.F-202603-063696

Timeline	Remediation Plan
Medium Term CAP-0074906	i. Relocate power receptacles from inappropriate locations to more suitable, protected areas to prevent exposure to environmental elements. For external receptacles, utilize suitably rated IP fixtures.

Medium Term CAP-0074907	ii. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk. Electrical Inspection
Medium Term CAP-0074908	iii. Replace or repair broken and exposed power receptacles immediately to restore safe and secure connections.
Medium Term CAP-0074909	iv. Relocate the power receptacle and install it at an appropriate, accessible location using a properly fixed mounting box, ensuring adequate ventilation, mechanical protection, and compliance with electrical safety standards.
Medium Term CAP-0076357	v. Remove the improper bulb holder, safely terminate all exposed conductors, and install an enclosed fitting with proper back box and conduit.
Longer Term CAP-0074910	i. Develop installation guidelines and training for personnel to ensure future power receptacles are installed in compliant and safe locations.
Longer Term CAP-0074911	ii. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.
Longer Term CAP-0074912	iii. Implement a preventive maintenance program that includes periodic checks and timely replacement of damaged electrical components to maintain safety standards.
Longer Term CAP-0074913	iv. Implement a regular inspection and maintenance schedule to check and secure all power receptacles, ensuring they remain safely and correctly installed.
Longer Term CAP-0076358	v. Develop facility-wide installation guidelines and training to ensure all lighting fixtures are properly enclosed with approved fittings, back boxes, and conduit per BS 7671.



Figure 2.77 - Improperly installed power receptacle on UPS in building 11



Figure 2.78 - Non-weather proof power receptacle installed outside the building 9



Figure 2.79 - Lighting bulb installed with wooden backing in the generator room

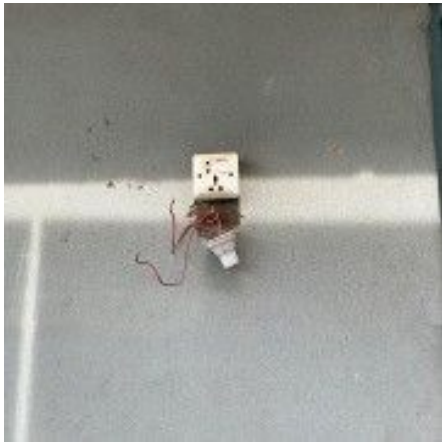


Figure 2.80 - Lighting bulb holder improperly connected in screen printing area in Garments Building



Figure 2.81 - Broken power receptacle found in building 3

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.